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Summary

B.Tech Computer Science Engineering graduate from VIT Vellore with a cross-domain technical profile spanning ML-based healthcare AI, IoT systems, and full-stack MERN development — including a live-deployed medication monitoring platform at senior-pill-final.netlify.app. Demonstrated operational and analytical thinking through internships at Tata Motors and Ardent Software Services. GMAT Focus Edition scorer with a structured interest in technology management, consulting, and MBA-oriented roles. Complemented by certifications from Wharton, Google, PwC, and Emory University in operations analytics, agile management, and data-driven decision-making.

Education

Vellore Institute of Technology (VIT Vellore)

B.Tech — Computer Science & Engineering / CGPA: **8.65 / 10**

Vellore, Tamil Nadu

Aug 2022 – Jul 2026

- **Relevant Coursework:** Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, Software Engineering, Compiler Design
- **Academic Highlights:** VITEEE All-India Rank 121 | 25% Merit Scholarship | **10.0 GPA** — Semester 6
- **National Academic Recognition:** Ranked **Top 1%** nationally — NPTEL Wildlife Ecology (2024) | **99% score** — NPTEL Forests & Their Management (2024)

St. Xavier's Institution

ICSE Class X | Academic Merit Scholarship for 90%+ in Class X

Kolkata, West Bengal

2020

Technical Skills

Languages: Java, C, C++, JavaScript, SQL, HTML5, CSS3

Frameworks & Libraries: React.js, Node.js, Express.js, Tailwind CSS

Databases: MongoDB, Firebase Realtime DB, SQLplus

Tools & Platforms: Git, GitHub, Postman, Google Colab, Microsoft Excel, Netlify, Vercel

IoT & Hardware: ESP32 Microcontroller, Sensor Integration, Firmware-Cloud Bridging

Concepts: DSA, OOP, DBMS, Federated Learning (FedAvg), IoT Systems, Workflow Optimization

Problem Solving: Practiced structured DSA problem solving in Java and C++ through academic coursework and self-learning

Core Competencies

Analytical Thinking | Strategic Problem Solving | Project Management | Operations Coordination | Stakeholder Communication | Data-Driven Decision Making | Cross-Functional Collaboration | Leadership | Presentation Skills | Process Optimization

Internship Experience

Tata Motors — Bhandari Alampur Commercial Vehicle Workshop

Kolkata, West Bengal

Operations & Service Analytics Intern

2026

- Analyzed **80+** vehicle service job cards and workshop records to map end-to-end service turnaround workflows across diagnostics, repair allocation, quality checks, and delivery stages.
- Identified **4 key operational bottlenecks** in technician scheduling and spare parts availability using Excel-based data structuring, contributing to a structured workflow improvement report.
- Proposed **6 targeted recommendations** for scheduling optimization and inventory coordination, aimed at reducing average vehicle downtime and improving service bay throughput.
- Documented operational findings in a structured analytics report covering workforce allocation patterns, inventory planning gaps, and process standardization recommendations.

Ardent Software Services

Kolkata, West Bengal

Full Stack Development Intern

Jun 2024 – Jul 2024

- Engineered a full-stack MERN Blood Bank Management System digitizing donor registration, emergency blood request workflows, and 3-tier administrative approval pipelines.
- Implemented RESTful API architecture with Express.js and MongoDB supporting role-based authentication across donor, patient, and administrator modules within a 6-week delivery window.
- Built a responsive React.js frontend with real-time request tracking dashboards, reducing multi-step coordination overhead across all three user roles.
- Received completion certification from Ardent Software Services upon successful delivery and evaluation of the full-stack application.

Future Hope NGO

Kolkata, West Bengal

Operations & Community Support Intern

2026

- Coordinated volunteer communication, donation inventory tracking, and resource distribution logistics across welfare drives involving **15–20 volunteers** and multiple community outreach activities.
- Supported operational execution and stakeholder coordination for NGO-led blood donation and blanket distribution campaigns serving underprivileged communities across Kolkata.

Key Projects

- IoT-Based Medication Monitoring System** | *React.js, Node.js, Firebase, ESP32, Tailwind CSS, Gmail SMTP*  Live Demo 
- Designed and deployed an integrated IoT medication adherence platform connecting ESP32 hardware sensors with Firebase real-time database, live at senior-pill-final.netlify.app — tested by **10+ users** including faculty and peers.
 - Automated caregiver alert workflows using Gmail SMTP, delivering missed-dose and low-stock notifications within **under 2 seconds** of trigger — eliminating manual monitoring overhead for caregivers.
 - Developed a centralized React.js caregiver dashboard with Tailwind CSS supporting **1 active patient profile** with real-time adherence tracking, medication scheduling, and cloud-synchronized inventory management.
 - Integrated firmware-cloud communication pipeline between ESP32 microcontroller and Firebase, enabling reliable real-time sensor data synchronization for medication dispensing events.
 - **Awarded S Grade** in VIT Vellore university evaluation; deployed on Netlify as a publicly accessible live application.
- Heart Disease Prediction using Federated Learning** | *Python, ML, Federated Learning, Google Colab*  GitHub
- Engineered a privacy-preserving healthcare prediction system using Federated Learning (FedAvg), enabling distributed model training across patient data nodes without exposing raw sensitive records.
 - Achieved **~95% F1-score** and **~90% accuracy** in centralized setup; maintained **~90% F1-score** and **~89% accuracy** in the federated configuration, validating minimal privacy-performance trade-off.
 - Compared centralized vs. federated architectures across classification metrics, demonstrating a production-viable privacy-preserving predictive workflow for healthcare AI deployments.
 - **Awarded S Grade** in B.Tech final-year project evaluation at VIT Vellore.
- Blood Bank Management System** | *React.js, Node.js, Express.js, MongoDB, SQL*  GitHub
- Developed a full-stack blood donation and emergency request platform supporting 3 user roles — donors, patients, and administrators — with structured approval workflows and centralized status tracking.
 - Implemented role-based authentication and multi-stage request handling pipelines, reducing coordination complexity during simultaneous emergency blood request scenarios.
 - Designed scalable RESTful API architecture enabling real-time request visibility and administrative oversight across all workflow stages.

Certifications & Professional Development

Operations Analytics 	The Wharton School, Univer-May 2026
Data-Driven Decision Making 	PwC via Coursera May 2026
Agile Project Management 	Google via Coursera May 2026
Consulting Approach to Problem Solving 	Emory University via Cours-May 2026
Product Management Essentials 	University of Maryland viaMay 2026
Career Skills in Data Analytics 	Coursera
	LinkedIn Learning (ID:May 2026
	29109d08)
Cloud-Based Full Stack Development with AI & ML (MERN)	Ardent Software Services Jul 2024
Certificate of Excellence: Java Fundamentals	Coding Ninjas 2025
Certificate of Excellence: HTML & CSS Fundamentals	Coding Ninjas 2025
Wildlife Ecology	NPTEL (Top 1% Nationally) Oct 2024
Forests and Their Management	NPTEL (Score: 99%) Apr 2024

Achievements, Awards & Competitive Exams

- **GMAT Focus Edition — Score 645 (2025):** Demonstrated quantitative and analytical reasoning ability as part of structured MBA preparation targeting top B-school programs.
- **VITEEE All-India Rank 121 (2022):** Secured AIR 121 among 200,000+ candidates nationally; awarded 25% merit-based tuition scholarship by VIT Vellore for academic excellence.
- **Perfect 10.0 GPA — Semester 6 (2025):** Achieved highest possible GPA in B.Tech Semester 6 CSE, reflecting consistent academic rigor and disciplined execution.
- **NPTEL Top 1% Nationally — Wildlife Ecology (2025):** Ranked among top 1% of all national NPTEL participants, earning Elite certification with distinction.
- **NPTEL 99% Score — Forests & Their Management (2024):** Scored 99% in a nationally administered NPTEL certification, demonstrating consistent high academic performance.
- **Coding Ninjas Certificate of Excellence (2025):** Recognized as a top performer in both Java Fundamentals and HTML & CSS Fundamentals among enrolled learners.
- **St. Xavier's Academic Merit Scholarship (2020):** Awarded merit scholarship for 90%+ performance in ICSE Class X.
- **Letters of Recommendation:** Received 3 faculty LORs from VIT Vellore recognizing analytical ability, project execution quality, and professional readiness.

Positions of Responsibility

Academic Peer Mentor — Core CSE Subjects	VIT Vellore
<i>Informal Academic Leadership Role</i>	<i>2022 – 2026</i>
– Mentored peers and junior students across DSA, OOP, DBMS, and OS through structured concept discussion sessions, supporting academic performance improvement among 5–10 students per semester. – Consistently approached as a go-to resource for project guidance, debugging support, and concept clarification within the CSE cohort over 4 years.	
Community Volunteer — Welfare Operations	Future Hope NGO, Kolkata
<i>Operational Support & Coordination Role</i>	<i>2024</i>
– Led volunteer team coordination and donation logistics across NGO-driven welfare programs, directly supporting blood donation drives, blanket distribution campaigns, and community outreach initiatives.	

Extra-Curricular Activities

- **Tabla — Classical Indian Percussion:** Sustained disciplined practice of Tabla over multiple years, developing structured commitment to skill-based learning, focus, and rhythmic precision.
- **Technical Presentations & Demonstrations:** Delivered project demonstrations and academic presentations on healthcare AI, IoT medication systems, and operations analytics before faculty panels and peer evaluators at VIT Vellore.
- **Community & Social Service:** Volunteered across multiple NGO-led welfare programs in Kolkata, contributing to blood donation drives and blanket distribution initiatives for underprivileged communities.
- **School-Level Cultural Participation:** Actively engaged in drama events, drawing competitions, and annual sports activities at St. Xavier’s Institution, reflecting consistent co-curricular involvement throughout school years.